Assignment No-1

1)

```
import java.util.*;
class Demo {
    public static void main (String [] args){
        Scanner sc = new Scanner (System.in);
        System.out.println ("enter a num:");
        int n = sc.nextInt ();
        for (int i=0; i<n ; i++) {
            for (int j=0 ; j<n ; j++){
                System.out.print ("* ");
            }
            System.out.println ( );
        }
    }
}
```

if: n = 4 j = 4

i = 0 n = 4

i<n = 0<4 → j = 0 < 4 — *

j = 1 < 4 — *

j = 2 < 4 — *

j = 3 < 4 — *

j = 4 < 4 — false, next line

i = 1 1<4 → j = 0 < 4 — *

j = 1 < 4 — *

j = 2 < 4 — *

$j = 3 < 4$

$\quad$ *

$j = 4 < 4$ — false

$i = 2 \quad 2 < 4 \longrightarrow j = 0 < 4$

$\quad$ *

$j = 1 < 4$

$\quad$ *

$j = 2 < 4$

$\quad$ *

$j = 3 < 4$

$\quad$ *

$j = 4 < 4$ — false

$i = 3 \quad 3 < 4 \longrightarrow j = 0 < 4$

$\quad$ *

$j = 1 < 4$

$\quad$ *

$j = 2 < 4$

$\quad$ *

$j = 3 < 4$

$\quad$ *

$j = 4 < 4$ — false,

$i = 4 \quad 4 < 4$ — false,

* * * * — Condition become false & go to next line

* * * *

* * * *

* * * *

2]

```
for (int i=1 ; i<=n ; i++) {
    for (int j=1 ; j<=n ; j++) {
        System.out.print( i + " ");
    }
    System.out.println ( );
}
```

if n=4

i=1 1<=4 — j=1<=4
└ print (i) = 1
j=2<=4
└ print (i) = 1
j=3<=4
└ 1
j=4<=4
└ 1
j=5<=4 — false, next line

i=2 2<=4 — j=1<=4
└ print (i) = 2
j=2<=4
└ 2
j=3<=4
└ 2
j=4<=4
└ 2
j=5<=4
└ false

i=3 3<=4 — j=1<=4
└ print (i)=3
j=2<=4
└ 3

j=3 <= 4
└ 3
j=4 <= 4
└ 3
j=5 <= 4 — false.

i=4 4<=4 — j=1<=4
└ print (i) = 4
j=2<=4
└ 4
j=3<=4
└ 4
j=4<=4
└ 4
j=5<=4 — false.

~~first~~ i=5 5<=4 – false, loop completed.

first iteration :	1	1	1	1
second	2	2	2	2
third	3	3	3	3
forth	4	4	4	4

3)

```
for (int i=1 ; i<=n ; i++){
    for (int j=1 ; j<=n ; j++) {
        System.out.print (j + " ");
    }
    System.out.println ();
}
```

if n=4

i=1 1<=4 — j=1<=4
└ print (j) — 1
j=2<=4
└ 2
j=3<=4
└ 3
j=4<=4
└ 4
j=5<=4 — false, next line

i=2 2<=4 — j=1<=4
└ 1
j=2<=4
└ 2
j=3<=4
└ 3
j=4<=4
└ 4
j=5<=4 — false, next line

i=3 3<=4 — j=1<=4
└ 1
j=2<=4
└ 2
j=3<=4
└ 3

j = 4 <= 4
4
j = 5 <= 4 — false. next line.

i = 4 4 <= 4 — j = 1 <= 4
1
j = 2 <= 4
2
j = 3 <= 4
3
j = 4 <= 4
4
j = 5 <= 4 — false.

i = 5 5 <= 4 — false
loop Completed.

first iteration	1	2	3	4	— next line
Second	1	2	3	4	
third	1	2	3	4	
forth	1	2	3	4	